

Drone Garage Updates

1/29/2026

To Order

N/A

Data Collection

- Metrics

- Throughput

- Methods

- Perhaps have some counter that updates on a receive and every second the receiver notes how many messages it got to measure like messages per second

- Network reliability

- What exactly?

- Perhaps measure how many were sent in a second and how many received in a second for each node pair?
 - Well we have only one drone sending messages out rn so we measure sends there and receives for all others then when we implement the relay messaging scheme we measure for each pair there?

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Throughput

- Basic stuff
 - Nothing too crazy
 - The example output below is for receiving
 - Note: the numbers are low because the heartbeat is like every second so just modify that
 - Will test higher heartbeat rate next slide
 - Images to show edits

```
92 void recieveMessageFromMesh(uint32_t from, String &msg) {
93     // Minimal logging to avoid serial slowdown
94     // Serial.printf("[%s] rx from %u: %u bytes\n", nodeName.c_str(), from, msg.length());
95
96     DynamicJsonDocument d(2048);
97     DeserializationError e = deserializeJson(d, msg);
98     if (e) { /* Serial.println("json err"); */ return; }
99
100     // Metric measuring stuff
101     messages_rec += 1;
102 }
```

```
223 // Metric measuring stuff
224 void printThroughputRec() {
225     Serial.printf("[%s] meshId=%u message throughput (received) = %u/s",
226         nodeName.c_str(), meshId, messages_rec);
227     messages_rec = 0;
228 }
```

```
234 // Metric measuring stuff
235 Task throughputRecTask(1000, TASK_FOREVER, &printThroughputRec); // every second print and reset
```

```
271 // Metric measuring stuff
272 userScheduler.addTask(throughputRecTask); throughputRecTask.enable();
```

```
[node4] meshId=2 message throughput (received) = 1/s[node4] meshId=2 leader=0 me? no lastHB=15041 now=16276
[node4] meshId=2 message throughput (received) = 0/s[node4] meshId=2 leader=0 me? no lastHB=17040 now=17276
[node4] meshId=2 message throughput (received) = 2/s[node4] meshId=2 leader=0 me? no lastHB=18041 now=18276
[node4] meshId=2 message throughput (received) = 1/s[node4] meshId=2 message throughput (received) = 1/s[node4] meshId=2 leader=0 me? no lastHB=19040 now=19276
[node4] meshId=2 leader=0 me? no lastHB=20041 now=20276
```

Throughput

- Higher heartbeat rate
 - Modifying the line “Task sendHeartBeatThroughMesh()” changing 1000 → 1 (2 node set up):
 - Note: node elected itself upon joining so will address that now
 - Temporary solution so it doesnt just re elect upon joining in first two images (with this temp solution, start node Ids at 1)
 - Third image first result
 - Fourth image second result
 - Eventually stops receiving heartbeats (prob not, prob just that sendHeartBeatThroughMesh() is sending a heartbeat every millisecond so just gonna tune it down)

```
217 void printStatus() {  
218     Serial.printf("[%s] meshId=%u leader=%u me? %s lastHB=%lu now=%lu\n",  
219         nodeName.c_str(), meshId, leaderId, amIGateway ? "yes" : "no",  
220         (unsigned long)lastHB, (unsigned long)millis());  
221     lastHB = millis(); // temporary  
222 }
```

```
151 void electGatewayCheck() {  
152     if (amIGateway) return;  
153     if (millis() - bootAt < DISCOVERY_MS) return;  
154  
155     bool newElection = (millis() - lastHB) > HBLOSS;  
156     static uint32_t prevLeader = 0;  
157     static bool     prevRole   = false;  
158  
159     if (newElection || (leaderId == 0)) { // temporary  
160         SimpleList<uint32_t> nodeList = mesh.getNodeList();
```

```
[node4] meshId=3 message throughput (received) = 4/s  
[node4] meshId=3 leader=1 me? no lastHB=9085 now=9277  
[node4] meshId=3 message throughput (received) = 5/s  
[node4] meshId=3 leader=1 me? no lastHB=10160 now=10277  
[node4] meshId=3 message throughput (received) = 5/s
```

```
[node4] meshId=3 leader=1 me? no lastHB=48276 now=49276  
[node4] meshId=3 leader=1 me? no lastHB=49276 now=50276  
[node4] meshId=3 message throughput (received) = 0/s  
[node4] meshId=3 leader=1 me? no lastHB=50276 now=51276  
[node4] meshId=3 message throughput (received) = 0/s  
[node4] meshId=3 leader=1 me? no lastHB=51276 now=52276  
[node4] meshId=3 message throughput (received) = 0/s
```

Throughput

- Tune down heartbeat rate
 - Modifying the line “Task
sendHeartBeatThroughMesh()” changing 1 → 50
(2 node set up):
 - Note, while uploading new code to the esp, on boot it started receiving again but the rate quickly went down to 0 but this is without updating the full set of esp
 - Receiving really is just 3~5 per second... anddddd drops to 0...
 - Ideas??? (not uploaded to the github yet just messing around)

```
[node4] meshId=3 message throughput (received) = 5/s
[node4] meshId=3 message throughput (received) = 5/s
[node4] meshId=3 leader=1 me? no lastHB=12182 now=12280
[node4] meshId=3 message throughput (received) = 4/s
[node4] meshId=3 leader=1 me? no lastHB=13011 now=13280
[node4] meshId=3 message throughput (received) = 4/s
[node4] meshId=3 leader=1 me? no lastHB=14109 now=14280
[node4] meshId=3 leader=1 me? no lastHB=15207 now=15280
[node4] meshId=3 message throughput (received) = 5/s
[node4] meshId=3 leader=1 me? no lastHB=16036 now=16280
[node4] meshId=3 message throughput (received) = 4/s
[node4] meshId=3 message throughput (received) = 3/s
[node4] meshId=3 leader=1 me? no lastHB=16860 now=17280
[node4] meshId=3 message throughput (received) = 0/s
[node4] meshId=3 leader=1 me? no lastHB=17280 now=18280
[node4] meshId=3 leader=1 me? no lastHB=18280 now=19280
[node4] meshId=3 message throughput (received) = 0/s
[node4] meshId=3 message throughput (received) = 0/s
```